CUSTOMER BEHAVIOR ANALYSIS

Internship Project Report

Submitted as part of the Data Analytics Internship Program

Submitted By

Vaishnavi Tiwari

Submitted To

Alfido Tech

Project Objective

To analyze customer purchasing behavior, perform customer segmentation using RFM analysis, identify churn trends, and provide data-driven recommendations to improve customer engagement and retention.

Tools Used

- Python
- Google Colab
- Pandas
- Matplotlib
- Data Visualization Techniques

Submission Date

29 June 2026

Introduction

Customer behavior analysis helps businesses understand how customers interact with products and services. By analyzing customer transactions, businesses can identify purchasing patterns, customer preferences, and factors that influence customer retention.

The objective of this project was to analyze customer purchase data, segment customers based on their behavior, identify churn trends, and provide recommendations to improve customer engagement and retention.

Dataset Description

The dataset contains customer transaction records along with demographic and behavioral information. It includes details such as customer age, gender, product category, purchase amount, payment method, returns, and churn status.

The dataset provides valuable information that can be used to understand customer purchasing habits and business performance.

Data Cleaning

Before performing the analysis, the dataset was cleaned to improve data quality and ensure accurate results.

The following steps were performed:

- Checked for missing values.
- Removed duplicate records.
- Converted the Purchase Date column into a proper date format.
- Validated transaction-related fields.
- Removed redundant information where necessary.

After cleaning, the dataset became more reliable and ready for further analysis.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500000 entries, 0 to 499999
Data columns (total 13 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   Customer ID            500000 non-null  int64
 1   Purchase Date          500000 non-null  object
 2   Product Category       500000 non-null  object
 3   Product Price          500000 non-null  int64
 4   Quantity               500000 non-null  int64
 5   Total Purchase Amount  500000 non-null  int64
 6   Payment Method         500000 non-null  object
 7   Customer Age           500000 non-null  int64
 8   Returns                405022 non-null  float64
 9   Customer Name          500000 non-null  object
 10  Age                    500000 non-null  int64
 11  Gender                 500000 non-null  object
 12  Churn                  500000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 49.6+ MB
```

Figure 1: Overview of the dataset structure, including features and data types

Feature Engineering

Additional features were created to gain deeper insights into customer behavior.

The following features were generated:

- Purchase Year
- Purchase Month
- Purchase Day
- Purchase Weekday
- Return Flag
- Customer Revenue

These new features helped in analyzing customer purchasing trends and revenue contribution more effectively.

Customer Segmentation Using RFM Analysis

RFM analysis was used to segment customers based on:

- Recency (how recently a customer made a purchase)
- Frequency (how often a customer purchases)
- Monetary Value (how much money a customer spends)

Based on the analysis, customers were grouped into the following segments:

Champions

These customers purchase frequently, spend more money, and have made recent purchases. They represent the most valuable customer group.

Loyal Customers

These customers regularly purchase products and contribute consistently to business revenue.

Potential Loyalists

These customers have shown recent activity and may become loyal customers with proper engagement strategies.

At-Risk Customers

These customers have not purchased recently and may stop engaging with the business if no retention efforts are made.

Key Findings

- Customer value varies across different segments.
- Champions generate the highest business value.
- At-Risk customers require targeted retention efforts.

- Potential Loyalists provide opportunities for future growth.

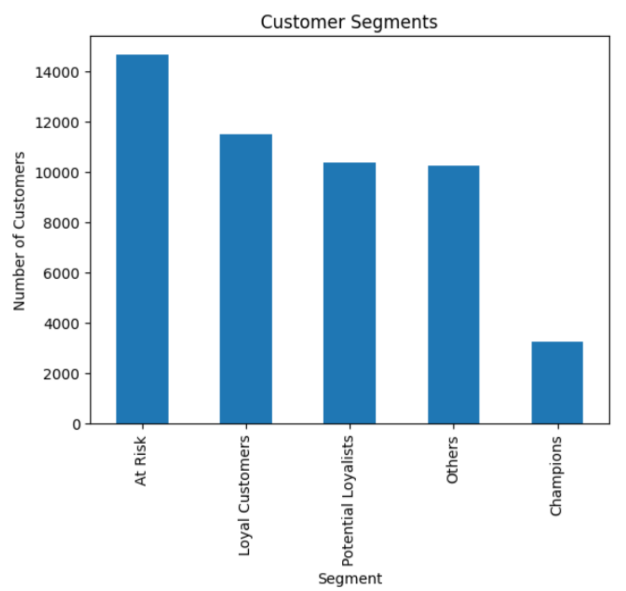

Figure 2: Customer segmentation based on RFM analysis.

Purchase Pattern Analysis

Several visualizations were created to understand customer purchasing behavior.

Monthly Purchase Trends

The analysis revealed fluctuations in customer purchases over time. Certain periods showed higher purchasing activity, indicating possible seasonal trends.

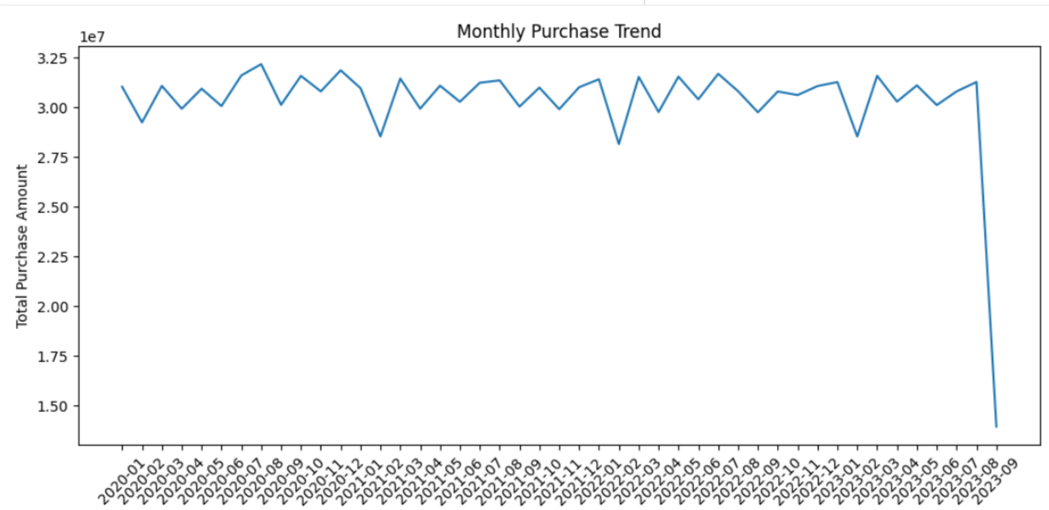

Figure 3: Monthly customer purchase trend.

Product Category Analysis

Some product categories generated significantly more revenue than others, indicating stronger customer demand for specific products.

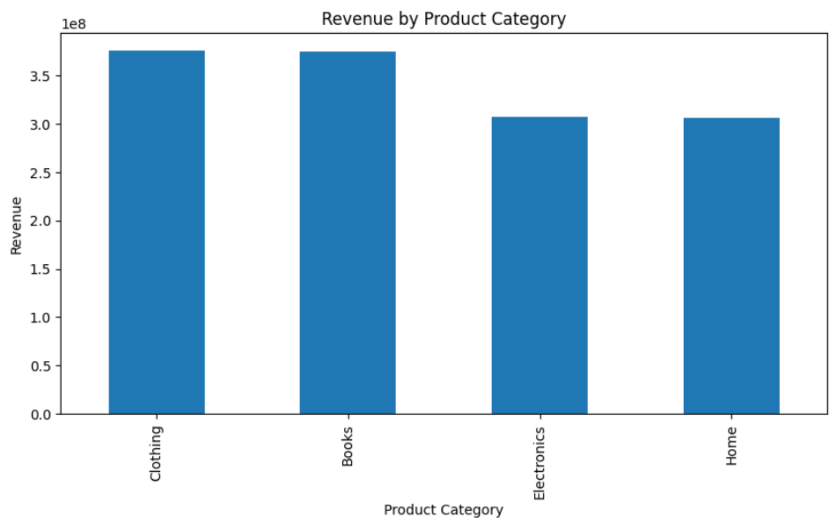

Figure 4: Revenue generated by different product categories.

Payment Method Analysis

Customers showed clear preferences for certain payment methods. Understanding these preferences can help improve the overall checkout experience.

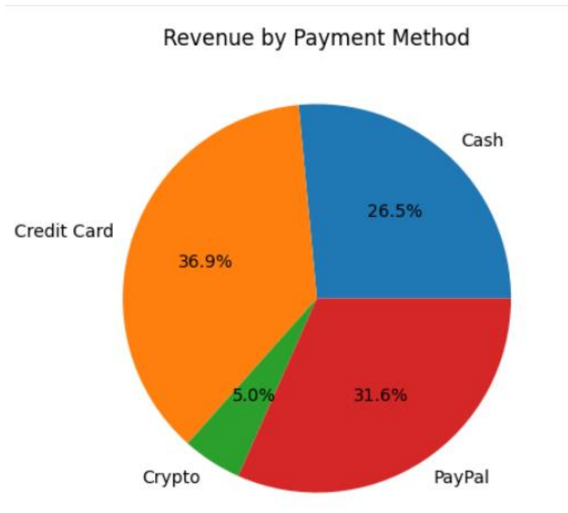

Figure 5: Distribution of revenue across payment methods.

Key Findings

- Customer demand varies across different product categories.
- Revenue is concentrated in specific categories.
- Purchase activity changes over time.
- Payment preferences influence customer purchasing behavior.

Churn Analysis

Customer churn analysis was performed to understand factors influencing customer retention.

Churn Distribution

The dataset contained both retained and churned customers, providing useful insights into customer retention patterns.

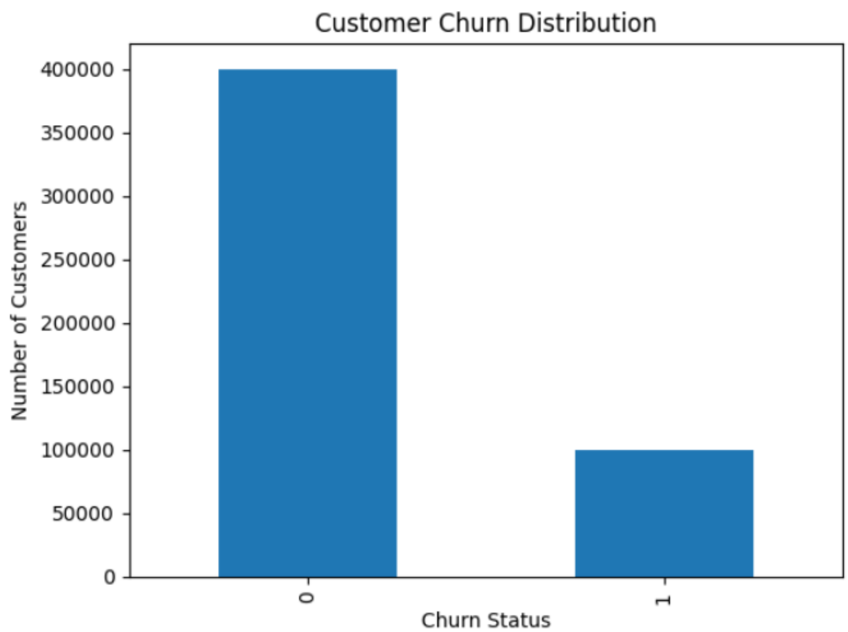

Figure 6: Distribution of retained and churned customers.

Churn by Product Category

Certain product categories experienced higher churn rates than others, suggesting potential differences in customer satisfaction or product performance.

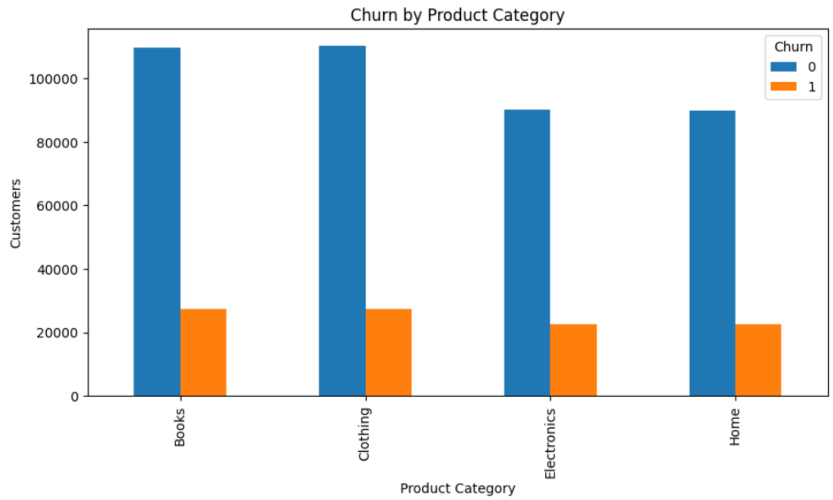

Figure 7: Churn distribution across product categories.

Churn by Age Group

Customer retention behavior varied across different age groups. Some age segments appeared more likely to remain active customers than others.

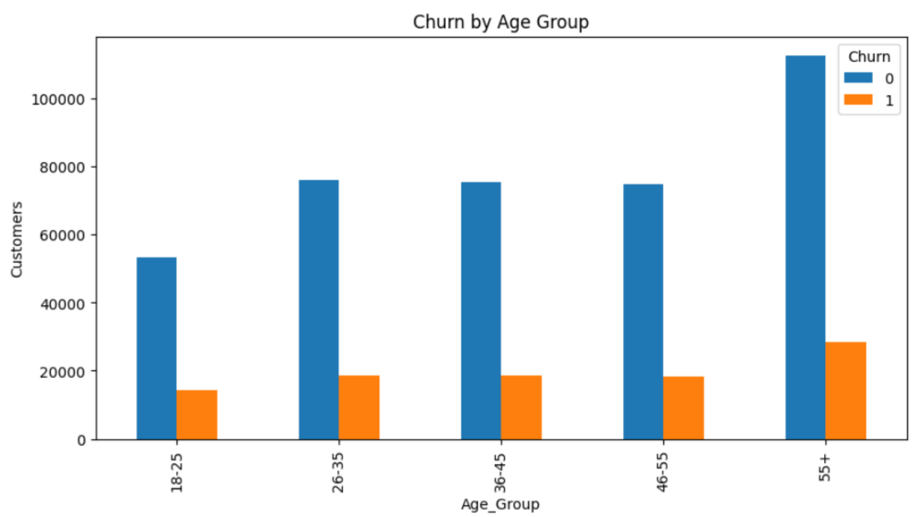

Figure 8: Churn behavior across different age groups

Returns vs Churn

Customers who returned products more frequently showed a higher tendency to churn. This suggests that customer satisfaction may play an important role in retention.

Key Findings

- Churn is influenced by multiple customer characteristics.
- Product returns may contribute to customer loss.
- Customer retention varies across age groups.
- Certain product categories require additional attention.

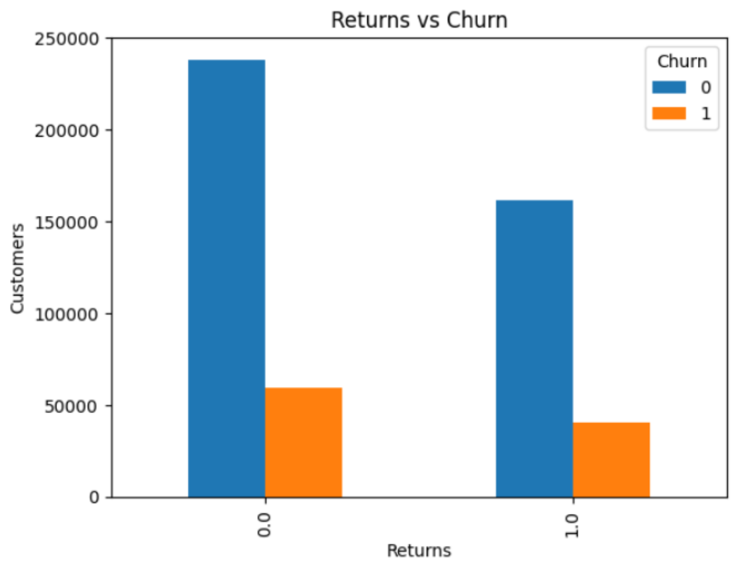

Figure 9: Relationship between product returns and customer churn.

Recommendations

Based on the customer segmentation, purchase pattern analysis, and churn analysis, the following recommendations are suggested to improve customer engagement and retention:

1. Launch Personalized Retention Campaigns

Customers identified as "At-Risk" should be targeted through personalized emails, discount coupons, and special offers. Re-engagement campaigns can encourage inactive customers to make repeat purchases and reduce churn.

2. Reward Loyal and High-Value Customers

Champions and Loyal Customers contribute significantly to overall revenue. Implementing loyalty programs, reward points, exclusive discounts, and early access to products can strengthen customer relationships and increase long-term engagement.

3. Improve Customer Experience in High-Churn Categories

Product categories with higher churn rates should be closely monitored. Customer feedback should be collected to identify issues related to product quality, pricing, or service, allowing the company to improve customer satisfaction.

4. Reduce Product Returns Through Better Support

The analysis suggests a relationship between product returns and customer churn. Providing accurate product descriptions, improving quality control, and offering responsive customer support can help reduce return rates and improve customer trust.

5. Use Segment-Based Marketing Strategies

Different customer segments exhibit different purchasing behaviors. Alfido Tech should implement targeted marketing campaigns based on customer age groups, purchasing history, and RFM segments to deliver more relevant offers and improve engagement.

Conclusion

This project analyzed customer transaction data to understand customer purchasing behavior, segment customers using RFM analysis, and identify churn patterns. Data cleaning and feature engineering were performed to prepare the dataset for analysis and improve data quality.

The RFM analysis successfully categorized customers into meaningful segments such as Champions, Loyal Customers, Potential Loyalists, and At-Risk Customers. Purchase pattern analysis revealed differences in customer spending behavior, product preferences, and payment method usage. Churn analysis highlighted factors that may influence customer retention, including age groups, product categories, and product return behavior.

Overall, the findings indicate that customer engagement can be significantly improved through personalized marketing, loyalty programs, enhanced customer support, and proactive retention strategies. By implementing the recommended actions, Alfido Tech can strengthen customer relationships, reduce churn, and improve long-term business performance.